Networking Basics: IP Address

An **IP (Internet Protocol) address** is a unique numerical identifier assigned to every device connected to a network. It allows devices to communicate with each other over the internet or a local network.

Why is an IP Address Important?

- Identifies a device on a network.
- Enables devices to send and receive data.
- Helps route information to the correct destination.

Types of IP Addresses

1. IPv4

- 32-bit address.
- Written as four decimal numbers separated by dots.
- Example: **192.168.1.10**
- Supports about **4.3 billion** unique addresses.

2. IPv6

- 128-bit address.
- Written as eight groups of hexadecimal numbers separated by colons.
- Example: **2001:0db8:85a3:0000:0000:8a2e:0370:7334**
- Supports an extremely large number of unique addresses.

Public vs. Private IP Address

Public IP Address	**Private IP Address**
Used on the internet	Used inside local networks
Assigned by an Internet Service Provider (ISP)	Assigned by a router or network administrator
Must be globally unique	Can be reused in different private networks
Example: 8.8.8.8	Example: 192.168.1.100

Static vs. Dynamic IP Address

Static IP

- Does not change.
- Manually assigned.
- Commonly used for servers, printers, and websites.

Dynamic IP

- Changes automatically.
- Assigned by a DHCP server.
- Commonly used for home computers and mobile devices.

Structure of an IPv4 Address

An IPv4 address has **4 octets**, each ranging from **0 to 255**.

Example:

192.168.1.10

- 192 → First octet
- 168 → Second octet
- 1 → Third octet
- 10 → Fourth octet

Common Private IPv4 Address Ranges

- **10.0.0.0 – 10.255.255.255**
- **172.16.0.0 – 172.31.255.255**
- **192.168.0.0 – 192.168.255.255**

Example

Suppose:

- Computer A IP: **192.168.1.5**
- Computer B IP: **192.168.1.10**

When Computer A sends data to Computer B, it uses Computer B's IP address to ensure the data reaches the correct device.

Key Points to Remember

- An IP address uniquely identifies a device on a network.
- IPv4 uses **32 bits**; IPv6 uses **128 bits**.
- Public IPs are used on the internet; private IPs are used within local networks.
- Static IPs remain the same; dynamic IPs are assigned automatically.
- Every device connected to a network needs an IP address to communicate.